Clinical Study Summary Report (CSSR) Document Status: Draft | **Version:** 0.1 | **Date:** 07-Apr-2026

---

1. Administrative & Study Identification

Full Study Title: A Phase III, Randomized, Open-label, Active-controlled, Multicenter Study Evaluating the Safety, Pharmacokinetics, Pharmacodynamic and Efficacy of Crovalimab Versus Eculizumab in Patients With Paroxysmal Nocturnal Hemoglobinuria (PNH) Currently Treated With Complement Inhibitors **Short Title/Acronym:** COMMODORE 1 (A Study Evaluating the Safety, Pharmacokinetics, and Efficacy of Crovalimab Versus Eculizumab in Participants With Paroxysmal Nocturnal Hemoglobinuria (PNH) Currently Treated With Complement Inhibitors) **Protocol Number:** BO42161 (Also: 2020-000597-26, 2023-506526-37-00) **NCT Number:** NCT04432584 **Posting ID:** 662341437294610788 **Sponsor/Company Name:** Roche (Hoffmann-La Roche) **Investigational Product:** Crovalimab (PiaSky®) **Active Comparator:** Eculizumab (Trade Name: Soliris | ATC Code: L04AA25 | EMA Product Information) **Phase of Development:** PHASE3 **Medical Monitor:** [Name and Contact Information]

---

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of crovalimab versus eculizumab in C5 inhibitor-experienced participants with Paroxysmal Nocturnal Hemoglobinuria (PNH). *(Note: The original primary study objective was efficacy; however, target recruitment was not met, and efficacy endpoints became exploratory with safety as the new primary objective).* **Secondary Objectives:** To evaluate pharmacokinetics (serum concentrations over time, immunogenicity) and exploratory efficacy endpoints including transfusion avoidance, hemolysis control, breakthrough hemolysis, hemoglobin stabilization, FACIT-Fatigue score, and patient preference.

2.2 Background & Rationale To address the burden of continuous intravenous infusions in PNH patients currently treated with complement inhibitors. Crovalimab is a novel C5 inhibitor designed to allow for low-volume, every-4-week, subcutaneous self-administration. This trial evaluates whether crovalimab offers a less burdensome, yet safe and effective, alternative to existing therapies (like eculizumab) for this lifelong disease. *Disease Background:* Paroxysmal nocturnal hemoglobinuria is a rare acquired hematologic disorder characterized by hemolytic anemia, dark-colored urine due to the release of hemoglobin in the blood, and thrombosis. The episodes of hemolysis tend to occur at night. It is caused by a somatic mutation in the glycosylphosphatidylinositol biosynthesis gene.

---

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active-controlled (Eculizumab) **Blinding:** Open-label **Randomization:** Randomized (1:1) **Trial Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting)

3.2 Planned Enrollment & Duration Targeted Enrollment: 190 participants **Number of Sites:** Global, Multicenter **Estimated Study Start:** 09-Sep-2020 **Estimated Primary Completion:** 24 weeks post-randomization (Primary Analysis), followed by an extension period up to 2+ years.

---

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Body weight $\ge$ 40 kg at screening (pediatric participants with body weight < 40 kg). 2. Treated with eculizumab or ravulizumab for PNH for at least 3 months prior to Day 1. 3. Lactate Dehydrogenase (LDH) levels $\le$ 2x the upper limit of normal (ULN) at screening. 4. Willingness and ability to comply with all study visits and procedures. 5. Documented diagnosis of PNH, confirmed by high sensitivity flow cytometry. 6. Vaccination against *Neisseria meningitidis* serotypes A, C, W, and Y < 3 years prior to initiation of study treatment; or, if not previously done, vaccination administered no later than one week after the first drug administration. 7. Women of childbearing potential: agreement to remain abstinent (refrain from heterosexual intercourse) or use contraception during the treatment period and for 10.5 months after the final dose of crovalimab or for 3 months after the final dose of eculizumab (or longer if required by the local product label).

4.2 Exclusion Criteria 1. History of allogeneic bone marrow transplantation. 2. History of myelodysplastic syndrome with Revised International Prognostic Scoring System (IPSS-R) prognostic risk categories of intermediate, high and very high. 3. Pregnant or breastfeeding, or intending to become pregnant during the study, within 10.5 months after the final dose of crovalimab, or 3 months after the final dose of eculizumab (or longer if required by the local product label). 4. Participation in another interventional treatment study with an investigational agent or use of any experimental therapy within 28 days of screening or within 5 half-lives of that investigational product, whichever was greater: participants enrolled in an eculizumab or ravulizumab interventional study are eligible provided they fulfill eligibility (e.g., are willing and able to comply with the study assessments) and stop their participation in current trial before randomisation/enrolment. 5. Positive for Active Hepatitis B and C infection (HBV/HCV). 6. Concurrent disease, treatment, procedure, or surgery or abnormality in clinical laboratory tests that could interfere with the conduct of the study, may pose any additional risk for the participant, or would, in the opinion of the investigator, preclude the participant's safe participation in and completion of the study. 7. History of or ongoing cryoglobulinemia at screening.

---

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Weight-based tiered dosing. Participants receive an intravenous (IV) infusion of crovalimab on Day 1, followed by subcutaneous (SC) injections once weekly from Day 2 through Week 4. From Week 5 onwards, crovalimab is administered via SC injection once every 4 weeks (Q4W). **Route of Administration:** Intravenous (Day 1) followed by Subcutaneous (SC) injections (potentially self-administered or by a caregiver). **Duration of Treatment:** 24-week primary treatment period, followed by an extension period (up to 96+ weeks).

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care therapies for PNH underlying conditions. **Prohibited:** Experimental therapies or participation in another interventional treatment study with an investigational agent within 28 days of screening or 5 half-lives, whichever is greater (except switching from an eculizumab/ravulizumab trial as specified).

---

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Flow Cytometry, LDH Labs
Baseline	Day 1	Randomization, First Dose (IV Infusion)
Interim	Weeks 1-4	Weekly SC injections, Safety Monitoring, Pharmacokinetic Sampling
Maintenance	Week 5 to 24	Q4W SC injections, Safety/ Efficacy Assessment, Education on self-administration
End of Primary Period	Week 24/25	Primary Safety Analysis, Secondary PK & Exploratory Efficacy Assessments

---

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes 1. **Percentage of Participants With Adverse Events (AEs) and by Severity:** Severity determined according to National Cancer Institute (NCI) Common Terminology Criteria for Adverse Events, Version 5 (CTCAE v5). 2. **Percentage of Participants With Injection-site Reactions, Infusion-related Reactions, Hypersensitivity and Infections** (including Meningococcal Meningitis). 3. **Percentage of Participants With Adverse Events (AEs) Leading to Study Drug Discontinuation.**

7.2 Secondary & Exploratory Outcomes 1. **Serum Concentrations of Crovalimab or Eculizumab Over Time.** 2. **Serum Concentrations of Ravulizumab at the Time of Crovalimab Initiation.** 3. **Percentage of Participants With Anti-crovalimab Antibodies.** 4. **Exploratory**

Efficacy: Proportion of participants achieving Transfusion Avoidance (TA) and Hemoglobin Stabilization; Hemolysis control (LDH $\le 1.5 \times$ ULN) and breakthrough hemolysis events; Patient preference (crovalimab vs. eculizumab) and FACIT-Fatigue score.

---

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring for AEs, injection-site/infusion-related reactions, and transient immune complex reactions (Type III hypersensitivity events) known to occur when switching between C5 inhibitors. Grading per NCI CTCAE v5. **Serious Adverse Events (SAEs):** Monitoring for life-threatening events, fatal AEs, and serious infections (e.g., meningococcal meningitis). **Data Monitoring Committee (DMC):** Independent internal/external committee details per Roche guidelines.

---

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all randomized participants who received at least one dose of the study drug). Efficacy analyses are exploratory. **Sample Size Justification:** Target enrollment was initially 190 but halted early due to the evolving treatment landscape; randomized patients were deemed sufficient for the revised safety primary endpoints. **Handling of Missing Data:** Standard imputation rules applied per sponsor protocol.

---

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standard onsite and remote monitoring visits per sponsor protocols. **Audit Trail:** eCRF locking and standard data clarification procedures.

---

11. Study Identifiers & Governance

Primary ID: BO42161 **Registry Number:** NCT04432584 **Posting ID:** 662341437294610788 **Sponsor/Company:** Roche **Official Regulatory Title:** A Phase III, Randomized, Open-label, Active-controlled, Multicenter Study Evaluating the Safety, Pharmacokinetics, Pharmacodynamic and Efficacy of Crovalimab Versus Eculizumab in Patients With Paroxysmal Nocturnal Hemoglobinuria (PNH) Currently Treated With Complement Inhibitors

---

12. Clinical Framework (PNH Specific)

Disease Area: Paroxysmal Nocturnal Hemoglobinuria (PNH) **Disease Ontology:** * **MeSH Code:** D006457 * **HPO Code:** HP:0004818 * **SNOMED ID:** 1963002 (Hierarchy: 323666000 | Anemia due to intrinsic red cell abnormality) **Diagnosis Threshold:** PNH confirmed by high sensitivity flow cytometry **Medical Methodology:** Complement C5 inhibitor switch therapy

---

13. Study Procedures & Methodology

Management Pathway: Transition from intravenous eculizumab/ ravulizumab to subcutaneous crovalimab **Education Protocol (HCP):** Administration guidelines for the loading IV dose and monitoring for transient immune complex reactions (TICRs). **Education Protocol (Patient):** Subcutaneous self-administration training (or caregiver training) commencing around Week 5/Week 9. **Quality Audit Mechanism:** Safety data reviews, PK/PD serum concentration validations, and patient preference questionnaires.

---

14. Observation & Follow-Up Schedule

Total Duration: 24-week primary phase + up to 96+ weeks extension period **Onsite Visit Cadence:** Baseline, Weeks 1-4, then Every 4 weeks (Q4W) for maintenance **Remote Monitoring Frequency:** As needed per study protocol **Checklist Review Interval:** Q4W in conjunction with SC dosing

---

15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	**[DD-MMM-YYYY]**
Clinical Operations Lead	**[Name]**	___	***[DD-MMM-YYYY]***
Lead Statistician	***[Name]***	_____	[DD-MMM-YYYY]